THE TWO BOOKS · Lecture 3 · the raw symbols

Alphabets & Codes — few units, vast output

Before meaning, before sequence — the symbols themselves

This lecture meets the smallest pieces of both Books: the letters of the Qur'an and the bases/codons/amino acids of the cell. The thesis starts here — a TINY alphabet, whose units carry NO meaning alone, generates an effectively unbounded lexicon. We prove the 'few → many' with real numbers from both worlds.

All Qur'an figures are computed from Book6; biology figures are mainstream, round. Every parallel is audited ✓ / ✗ / ~.

The Two Books — two revelations of one Author

عالم التدوين — the WORD (قول الله)

The Qur'an: God's speech in language. The Book of SCRIPTURE — tadwīn.

عالم التكوين — the ACT (فعل الله)

The living cell: God's deed in creation. The Book of CREATION — takwīn.

One Author · one reader

Same source (Allah); primary addressee the human (insān). Both are āyāt. Here we compare their ALPHABETS.

Two small alphabets, side by side

the raw symbols — neither carries meaning on its own

QUR'AN — 28 letters
ا ب ت ث ج ح خ د … ي

GENOME — 4 bases
A · T · G · C

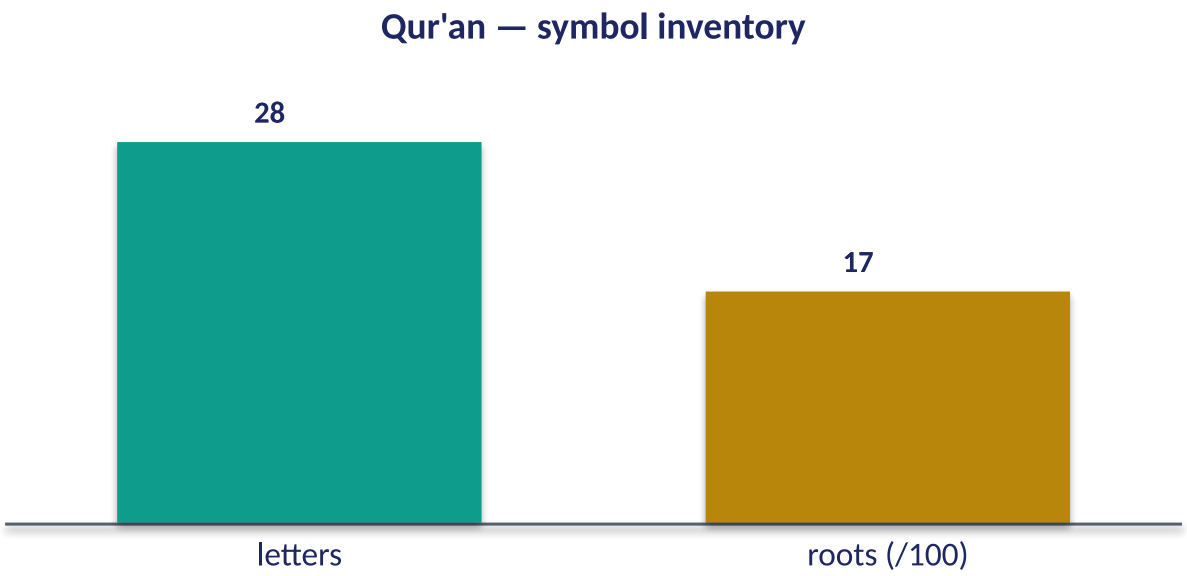
→ 1,702 roots
the Qur'anic 'code-book'

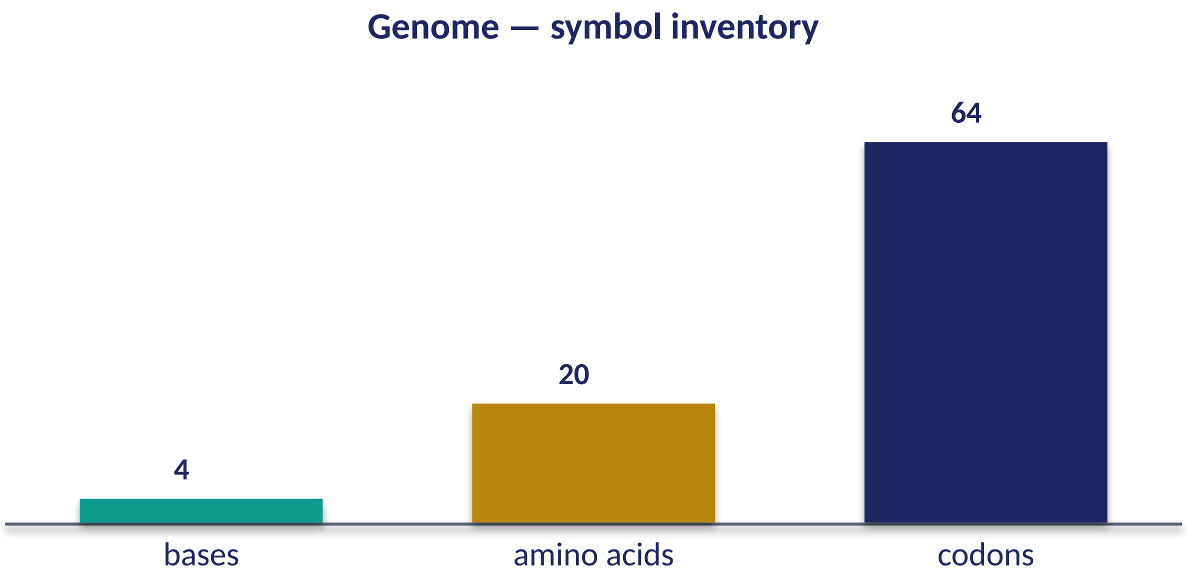
→ 64 codons → 20 amino acids
the cell's code-book

A letter is not a word; a base is not a protein

Each system starts from a closed, tiny set of meaningless tokens and a small derived 'code-book'. Everything else — every word, every protein — is BUILT from these. The rest of the lecture asks: how far can so few go?

Real data — how small the alphabets really are





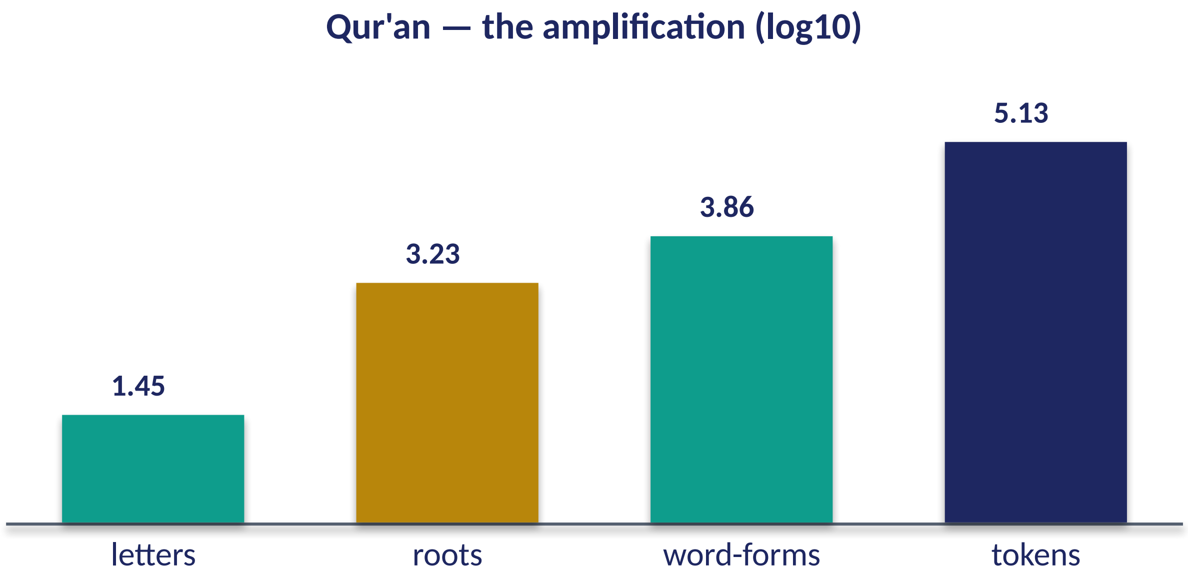
28 letters, a triplet code-book

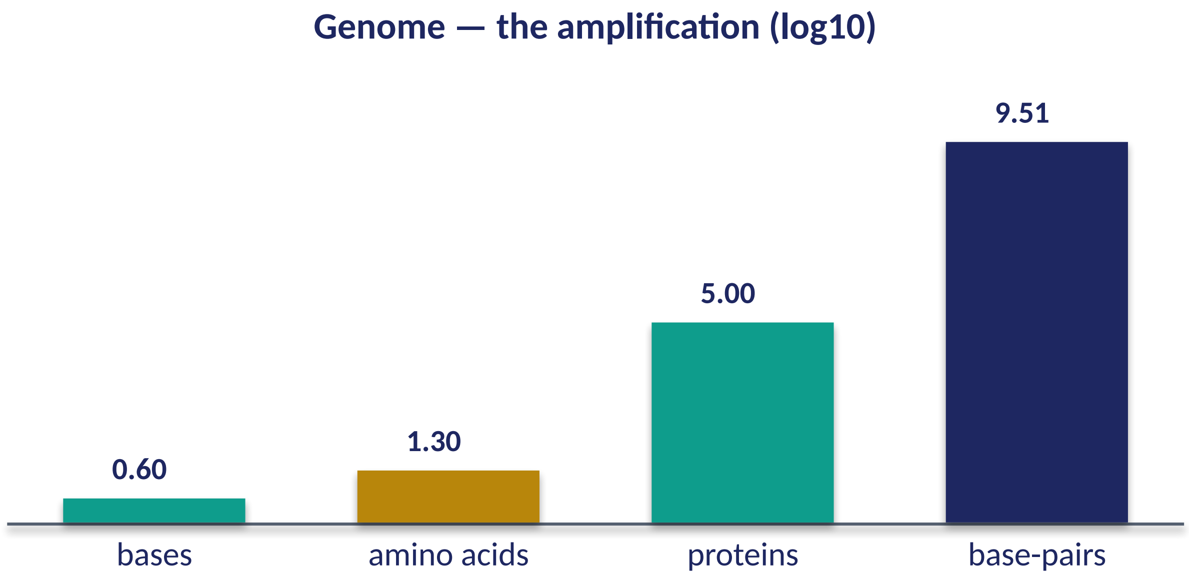
28 written letters generate ~1,702 roots — and ~96% of those roots are exactly THREE letters. A triplet code over a tiny alphabet.

4 bases, read in threes

4 bases form $4^3 = 64$ codons, which specify 20 amino acids. The cell, too, builds everything from a handful of symbols read three at a time.

PILLAR 1 — few units, vast output (log10 count)





28 letters → 7,236 distinct words

Computed from Book6: 28 letters → 1,702 roots → 7,236 distinct word-forms → 135,366 word-tokens. Each single letter ultimately generates ~268 distinct words.

20 amino acids → ~100,000 proteins

4 bases → 64 codons → 20 amino acids → ~80–100k proteins, from ~3.2 billion base-pairs. A tiny alphabet, an effectively unbounded proteome. Same shape, both Books.

The same funnel runs in both Books

a few symbols at the top, an unbounded lexicon at the bottom

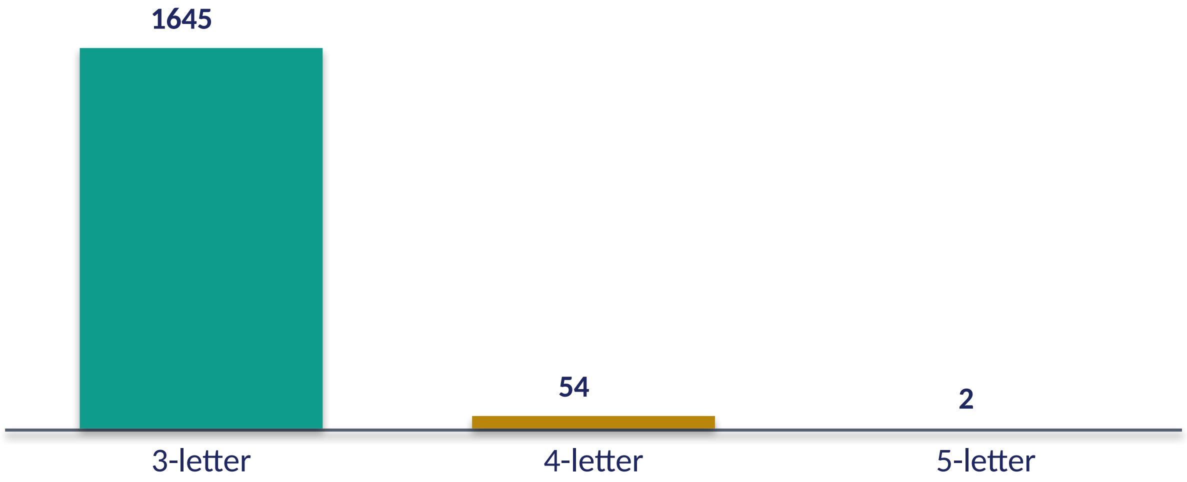
28 letters	4 bases
1,702 roots	20 amino acids
7,236 word-forms	~100k proteins
135,366 tokens	3.2 Gbp

Generativity is the shared signature

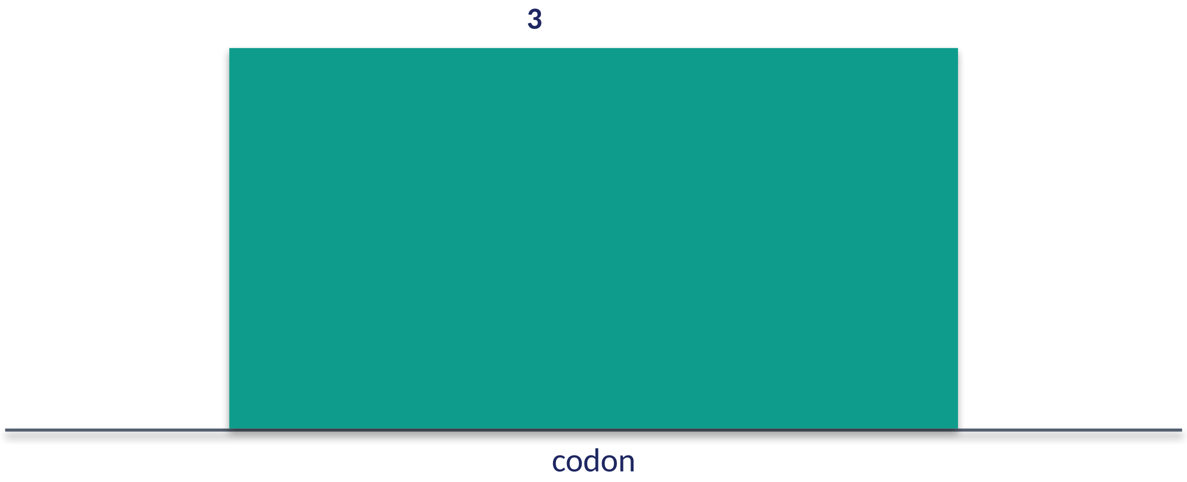
Both narrow at the top to a closed symbol set and widen without bound at the bottom. The engine of the widening — order and expression — is the rest of the course.

Real data — both codes are read in THREES

Qur'an — root length (count of roots)



Genome — codon length (bases per codon)



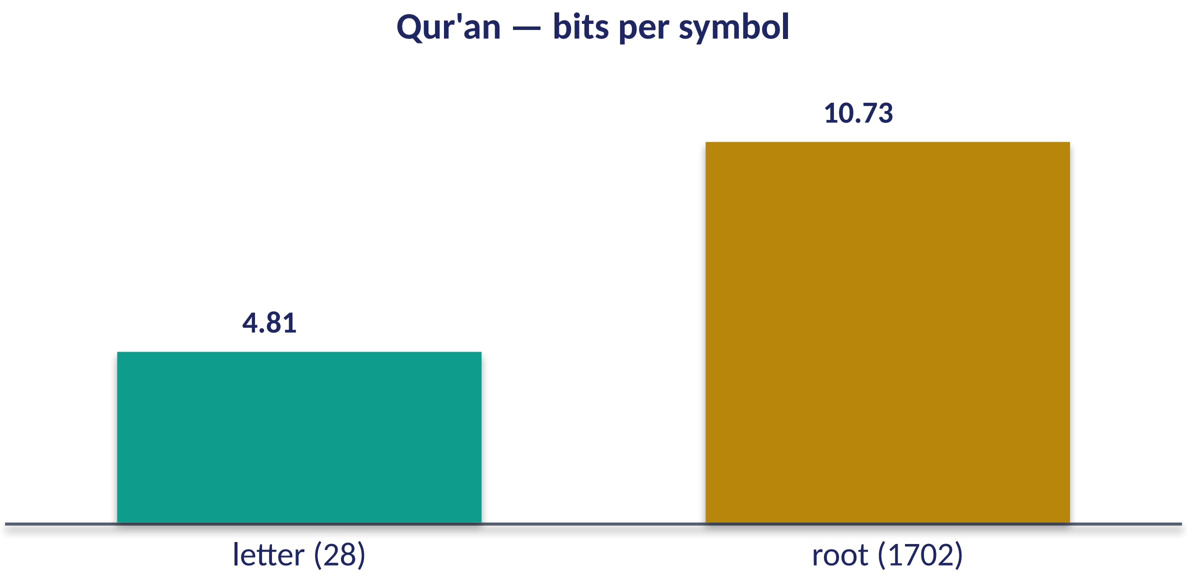
~96% of roots are triliteral

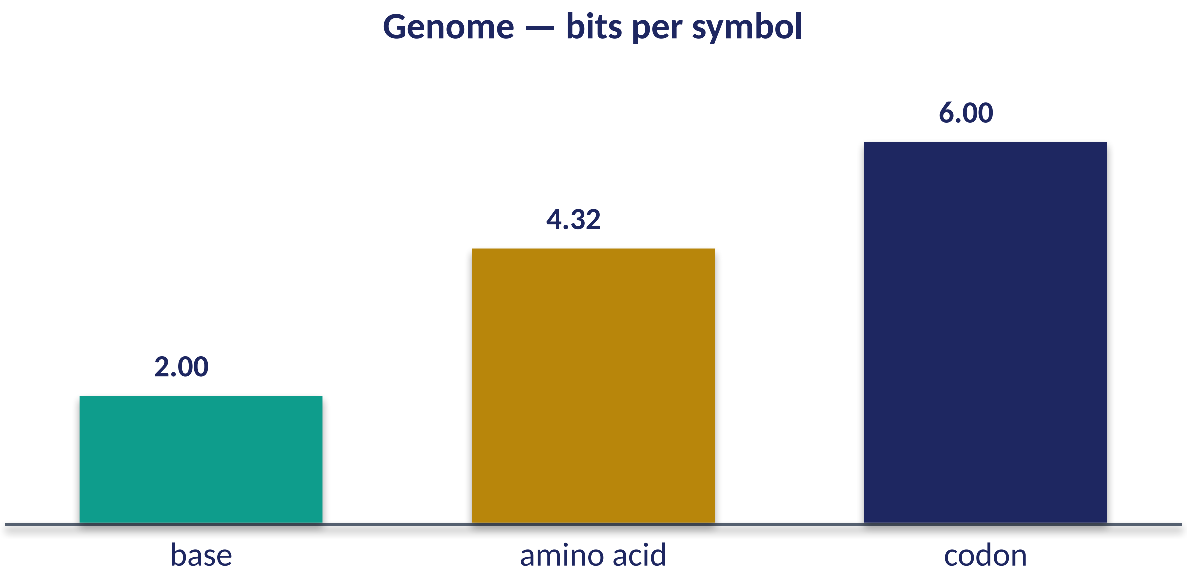
1,645 of 1,702 roots are exactly 3 letters. Arabic morphology is overwhelmingly a TRIPLET system — a near-universal triconsonantal skeleton.

Every codon is a triplet

The genetic code reads bases in non-overlapping groups of three: $4^3 = 64$ codons. Two independent Books converged on a triplet unit — a structural echo worth noting.

Real data — how much each symbol carries (bits)





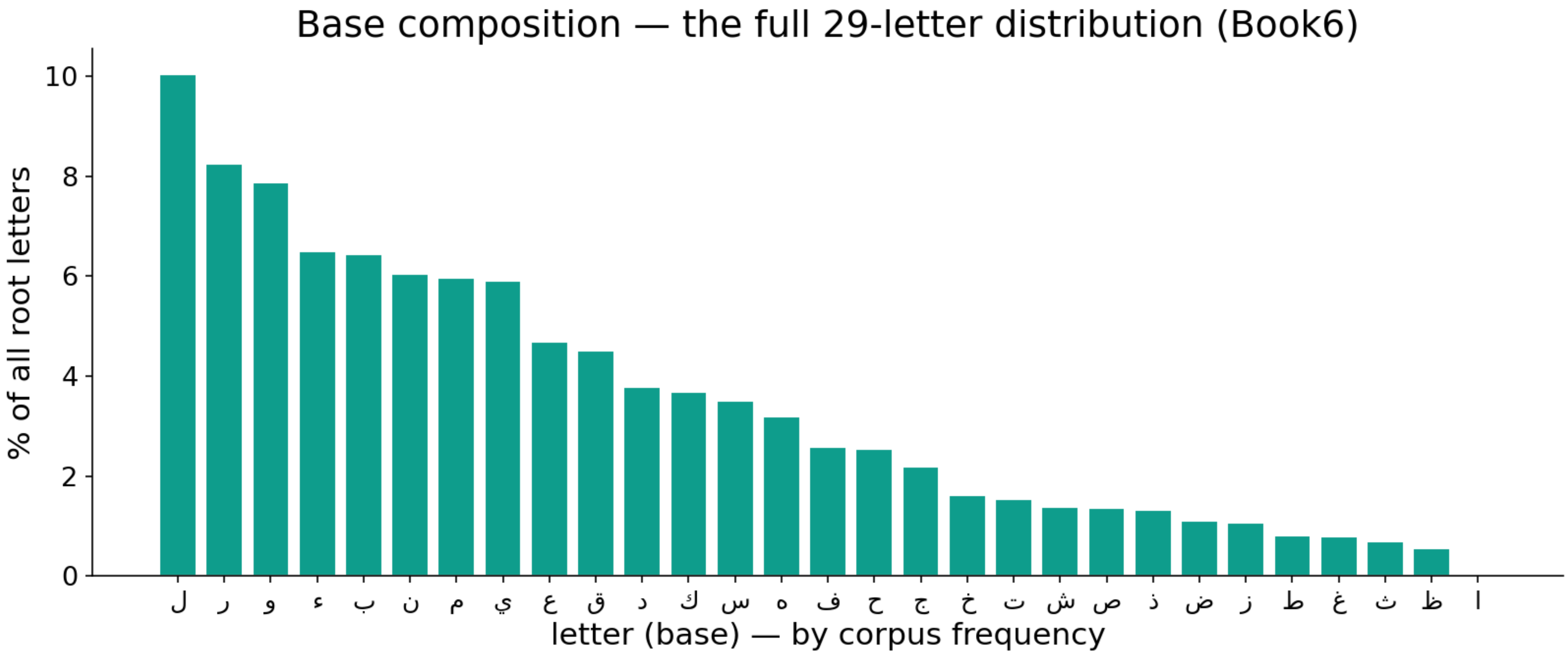
Small alphabets, exact capacity

A letter (28 options) carries log2(28)=4.8 bits; a root ~10.7. The information a symbol holds is fixed by the alphabet size — measurable, not mystical.

A base = 2 bits, a codon = 6

log2(4)=2 bits per base; a codon 6; an amino acid log2(20)=4.3. Both Books pack meaning into a handful of low-bit symbols. Information, not magic.

Real data — the full letter (base) distribution



In the data — every letter's share of all root-letters (Book6): ا 18%, ل 11.5% … a long tail across 28 symbols, far from the 3.6%-each of a uniform alphabet. The proteome is likewise skewed (Leu ~9.7% → Trp ~1.1%) — both Books have a characteristic, non-random composition.

PILLAR 2 preview — the unit is empty, ORDER speaks

the SAME three letters ب ح ر — six different roots

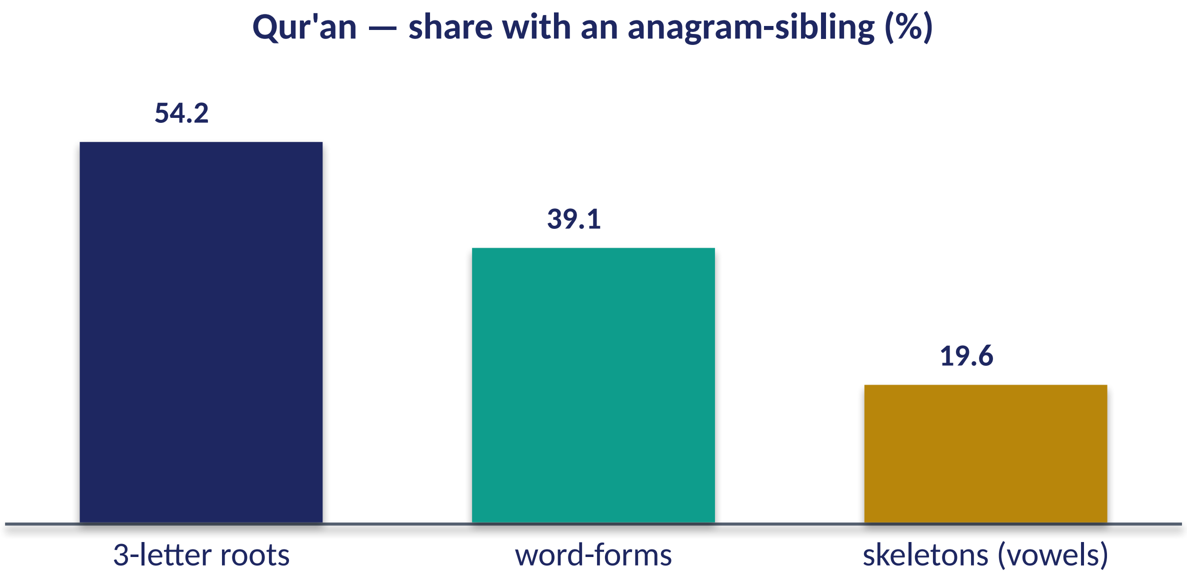
بحر	برح	حبر	حرب	ربح	رحب
sea	depart	ink	war	profit	ease

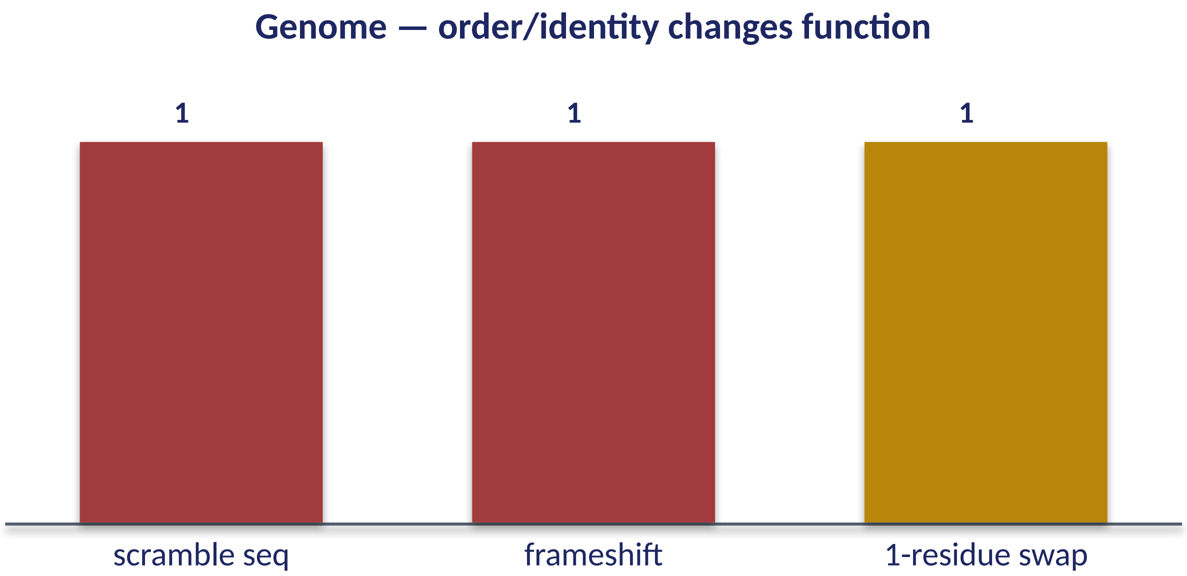
Three meaningless consonants, six unrelated meanings

None of ب, ح, ر means anything alone. Their ORDER alone separates 'sea' from 'war' from 'profit'. The biological mirror: the same bases read in a shifted frame (frameshift), or the same amino acids in a scrambled order, give an entirely different — usually non-functional — protein.

The unit carries no meaning. The SEQUENCE carries all of it. (Proved at scale in Lecture 5.)

Real data — order matters across the whole lexicon





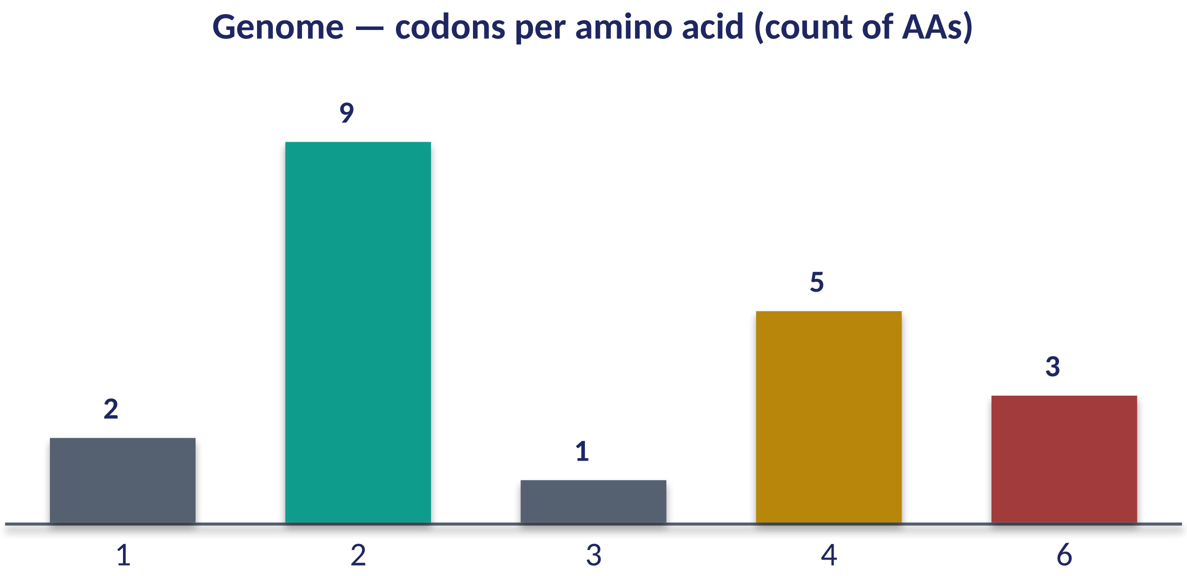
Reordering is everywhere

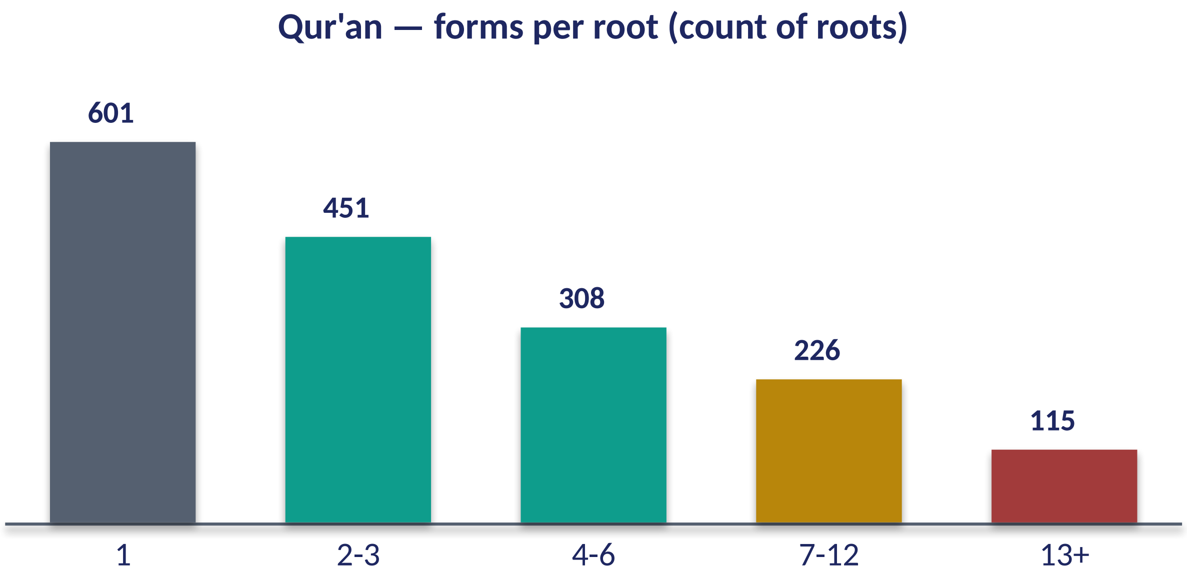
54% of 3-letter roots share their letter-set with another root; 39% of word-forms have an anagram-sibling; 19.6% of skeletons carry ≥2 meanings via vowels alone. Order, not inventory, makes the word.

Same parts, different order, new function

Scramble a protein's residues → no fold; shift the reading frame → new protein; swap ONE residue (sickle-cell, Glu→Val) → disease. In the cell too, arrangement is function.

Real data — redundancy in the code-book





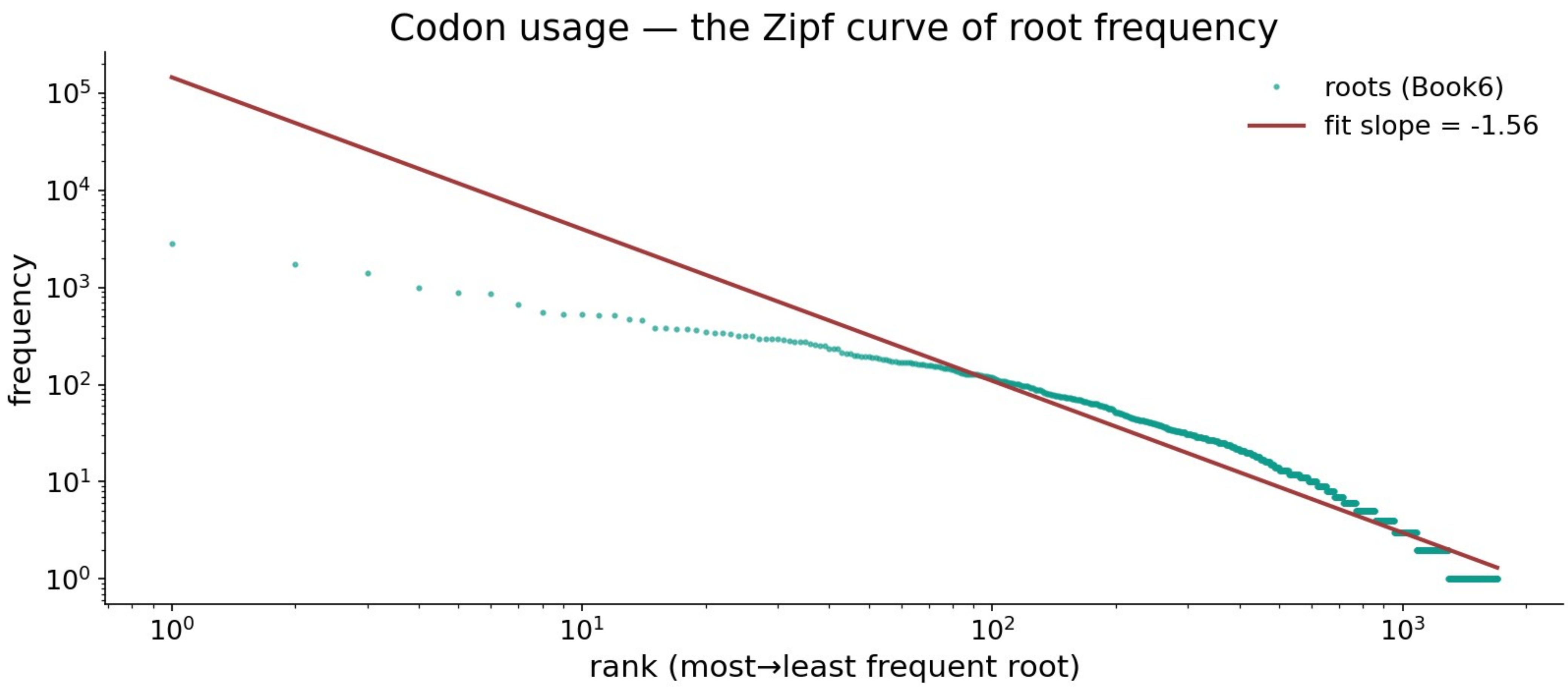
Many codons → one amino acid

18 of 20 amino acids have 2–6 codons (only Met, Trp have one). Redundancy buffers copying errors — many spellings, one meaning.

One root → many forms

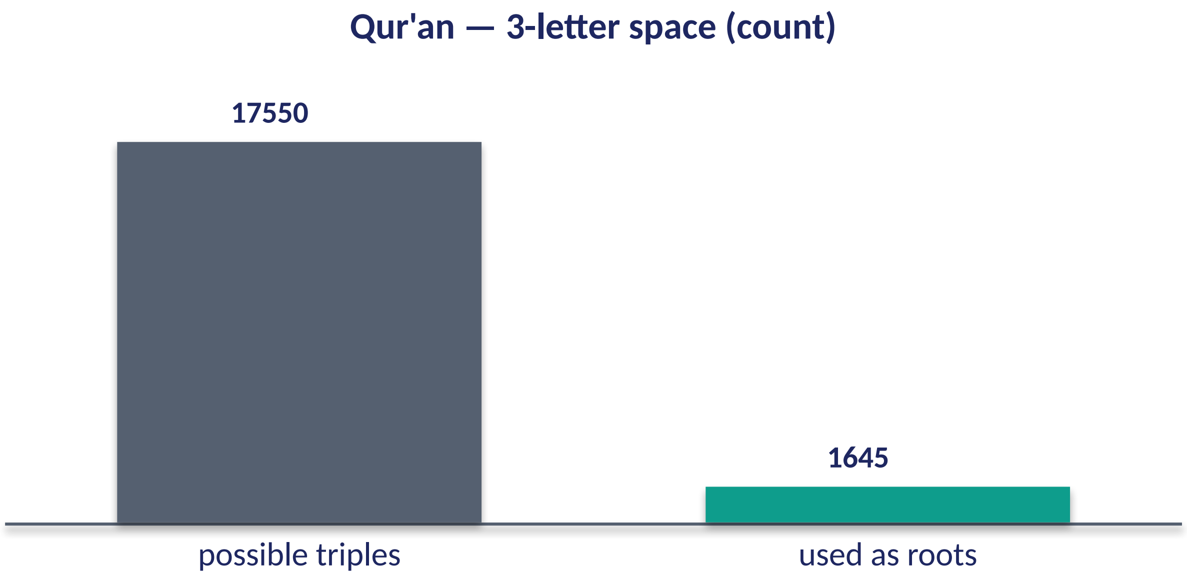
Conversely, a single root fans into many surface forms (mean 4.7, max 648). Both Books exploit a many-to-one and one-to-many slack between symbol and output.

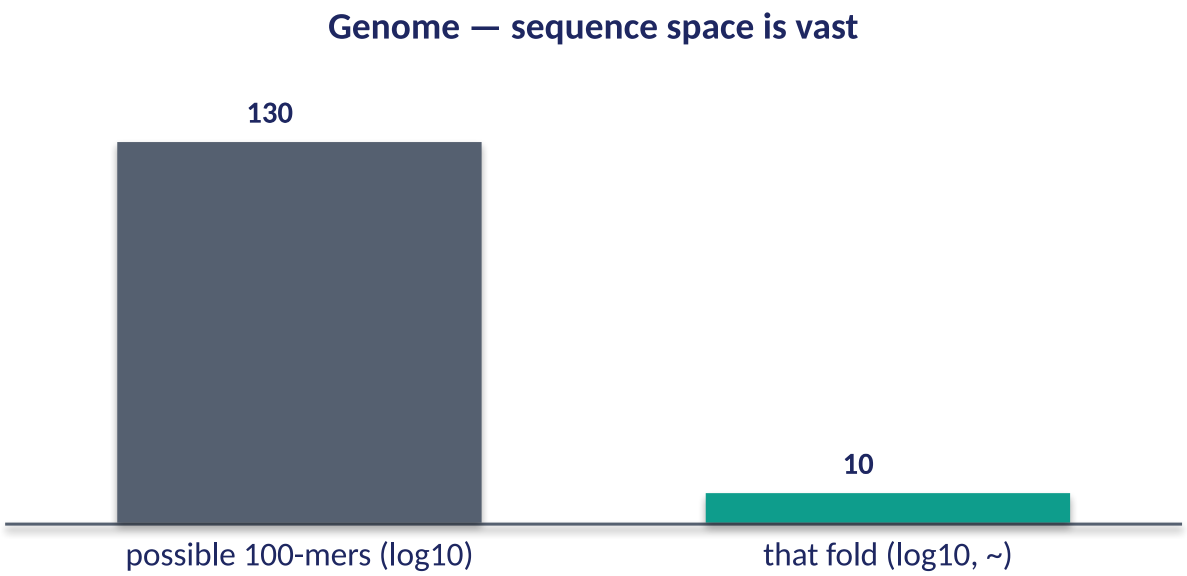
Real data — root usage is heavy-tailed (the Zipf curve)



In the data — all 1,700 roots ranked by frequency on log-log axes; the fitted slope ≈ −1.56 — 1722 قول, 2851 ءله then a long power-law tail. Amino-acid usage shows the same heavy-tailed economy (Leu/Ser common, Cys/Trp rare).

PILLAR 3 preview — most of what's possible is unused





Only 9.4% of triples are realized

Of 17,550 possible ordered 3-letter combinations, the Qur'an uses 1,645 as roots — 9.4%. The vast majority of the space is silent, never expressed.

A vanishing fraction of sequences fold

A 100-residue chain has ~10^{130} possible sequences; only an astronomically tiny fraction fold into working proteins. In both Books, POSSIBILITY ≫ realization — expression selects. (Lectures 9, 12.)

Audit — supported, broken, silent

✓ SUPPORTED

Both Books build unbounded output from a tiny, closed, non-uniform alphabet read in threes; order distinguishes units; most of the possible space is unused. All measured, both worlds.

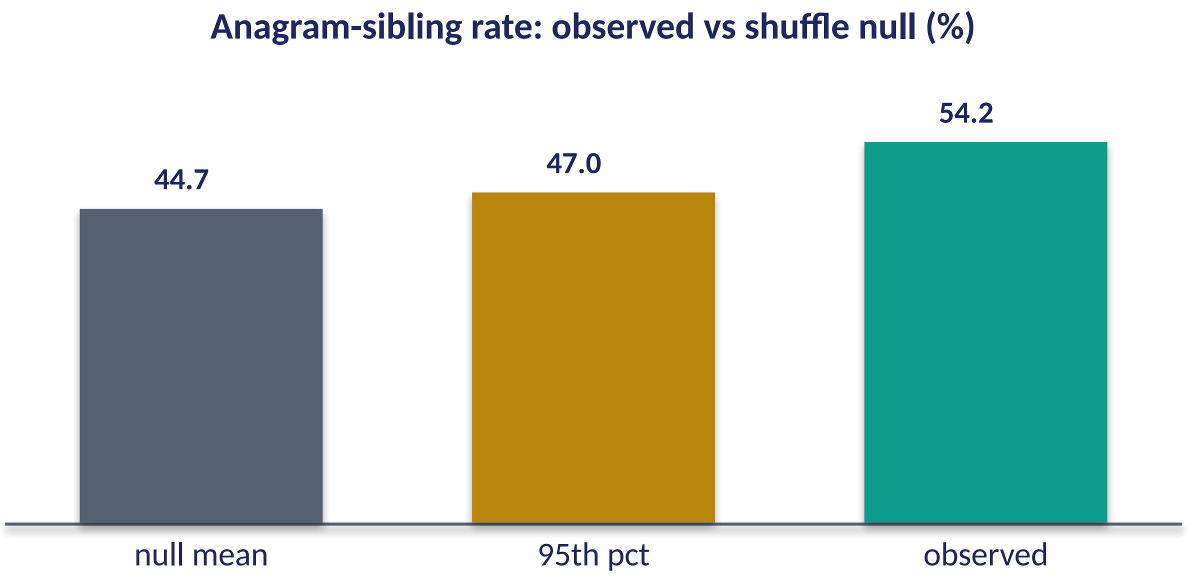
✗ BREAKS

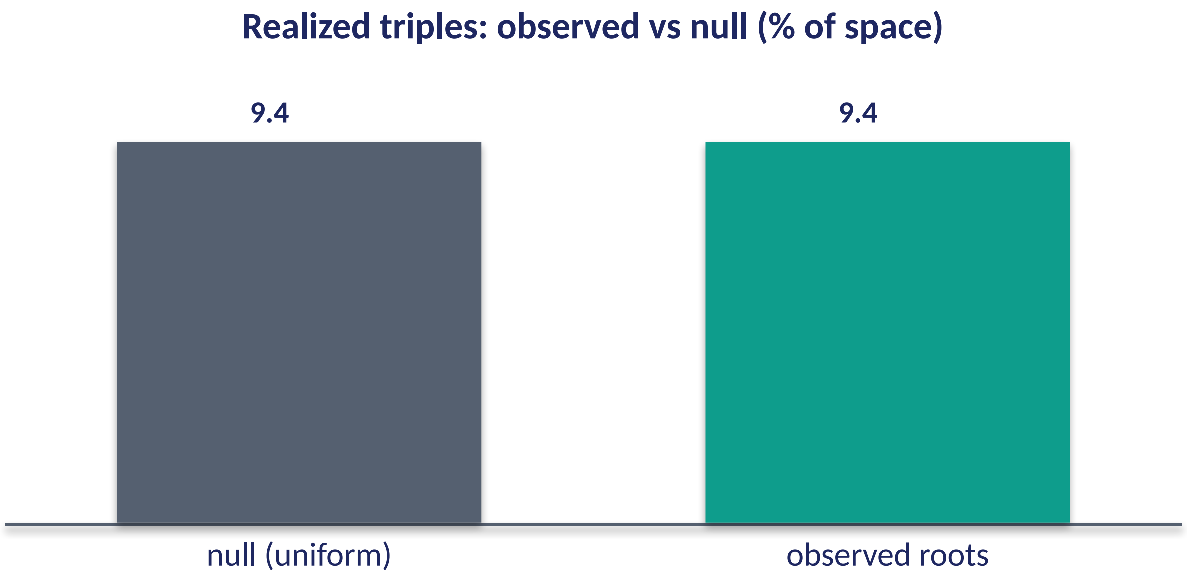
The triplet 'echo' is not identity: a codon (3 bases→1 amino acid) is not a root (3 letters→a concept). Similar STRUCTURE, different substance. No base ↔ letter mapping is claimed.

~ SILENT (surmisable)

Whether the shared 'triplet + heavy tail + sparsity' profile reflects a deep constraint or convergent economy is open — interesting, untested, not asserted.

Validation — is 'order matters' more than chance?





The anagram rate is real, not chance

A frequency-matched null gives ~45% anagram-siblings by chance (95th pct 47%); the Qur'an's ~54% sits well past it ($p<0.003$) — Arabic actively REUSES letter-sets in different orders. Order is doing real work.

Sparsity is structured, not random

The realized roots are not a random 9.4% of the space — they cluster around pronounceable, productive triconsonantal skeletons. Expression is selective, and we can measure it.

What this lecture established — the thesis, in data

the spine of the whole course, proved from the symbols up

① FEW → MANY
28 letters → 7,236 words ; 20 AA → ~100k proteins

② ORDER SPEAKS
6 → ب ح ر roots ; 54% anagram rate

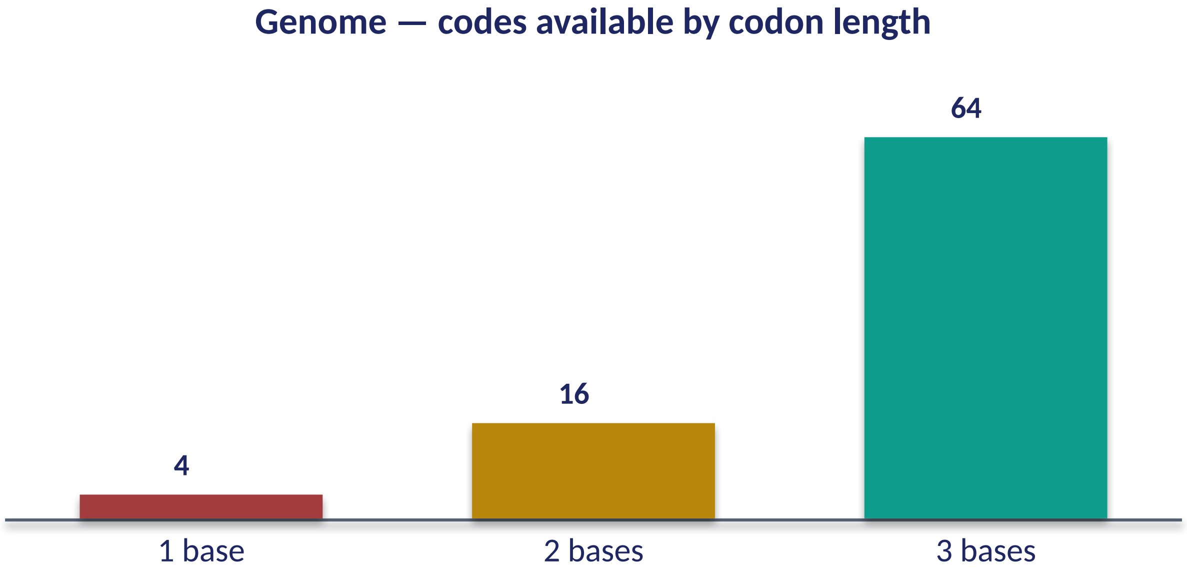
③ EXPRESSION
9.4% of triples used ; mean 4.7 forms/root

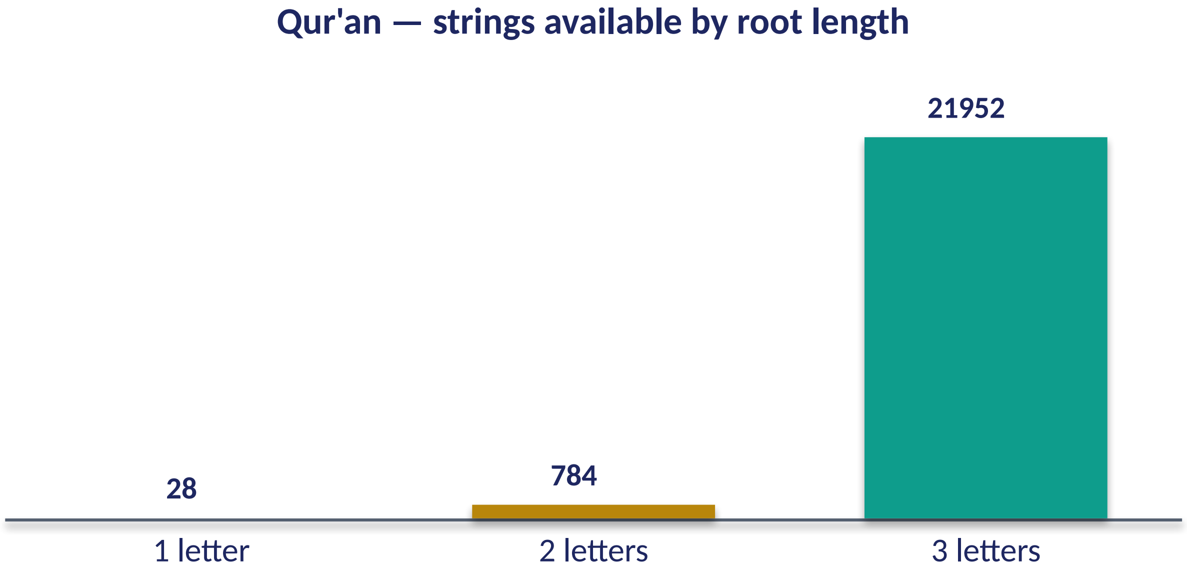
The unit is meaningless; the system is generative

From a tiny, non-uniform, triplet alphabet — whose individual symbols mean nothing — both Books generate an unbounded lexicon. ORDER turns the same symbols into different meanings; EXPRESSION decides which of the astronomically-many possibilities actually exist.

Pillar 1 is proved here. Pillar 2 (order) is the work of Lectures 5–6; Pillar 3 (expression) of Lectures 9 and 12.

Real data — why THREE is the minimum that works





2 bases cannot name 20 amino acids

One base gives 4 codes, two give 16 — both short of the 20 amino acids needed. Three bases give 64: the SMALLEST length that suffices. The triplet codon is forced by capacity.

2 letters cannot index 1,702 roots

28 letters: one gives 28, two give 784 — short of 1,702 roots. Three give 21,952: enough, with huge headroom. That headroom is exactly the 9.4% sparsity — capacity to spare, mostly unexpressed.

Synthesis & discussion

THE BIG IDEA

Generativity is the deepest thing the two Books share at the symbol level: a closed, tiny alphabet whose meaningless units, through order and selective expression, build an open-ended lexicon. We measured every claim in both worlds — no parallel was assumed.

FOR DISCUSSION

• Why do two independent Books both read in threes — constraint, or coincidence? • If a letter means nothing, where does meaning 'live'? • Is 9.4% realized a sign of design, economy, or just pronounceability? • What would falsify 'order matters'?

Real-world relevance & takeaway

REAL-WORLD RELEVANCE

'Few symbols → unbounded output by arrangement' is the principle behind language, the genetic code, digital data, and music. Seeing it in scripture AND the cell trains the eye for generative systems everywhere.

THE TAKEAWAY

A tiny alphabet of meaningless units builds an unbounded lexicon — because ORDER distinguishes them and EXPRESSION selects them. Proved with real data, both Books.